Haider Ali

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Research Interests

Remote Sensing, Image Super-Resolution, Generative (Diffusion) Models, Computer Vision, and Trustworthy Uncertainty-Aware Geospatial AI.

Education

National University of Sciences and Technology (NUST), Pakistan, MS in Artificial Intelligence (CGPA: 3.55/4.0) Aug 2022 – Apr 2025

Relevant Courses: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Artificial Intelligence, Mathematical Methods for Artificial Intelligence
Thesis: Offline Handwritten Signature Generation Using Generative Models (GPA: 4.0/4.0)
Supervisor: Prof. Dr. Faisal Shafait

Balochistan University of IT, Engineering and Management Sciences (BUITEMS), Pakistan, BS in Software Engineering (CGPA: 3.92/4.0) Aug 2017 – Dec 2021

Dean's Honor List

English Proficiency: IELTS Academic, overall band 7.0

Publications

Haider Ali, Maxim Dolgich, Ulrich Schwanecke, Adrian Ulges, Faisal Shafait. **Non-local sparse attention based multi-image super resolution**. [Accepted IGARSS 2026 (Poster), Washington, D.C., USA] [Paper][Code]

Haider Ali, Momina Moetesum, Adnan Ul-Hasan, Faisal Shafait, M. Imran Malik. **GeoCT-SigNet: A Geometry-Aware Hybrid CNN–Transformer Network for Offline Signature Verification**. [Under review in IEEE Access 2026] [Preprint][Code]

Haider Ali, Maxim Dolgich, Fabian Stahl, Ulrich Schwanecke, Adrian Ulges, Faisal Shafait. **RGB Super-Resolution for Multispectral Remote Sensing Images**. [Accepted IGARSS 2025 (Poster), Brisbane, Australia][Paper][Code]

Research Experience

Research Assistant, Deep Learning Lab (DLL), NCAI, NUST, Pakistan Jan 2026 – Ongoing

- Developing cloud-robust SAR–optical semantic segmentation trained on clear-sky images only: +16.0 mIoU on real thick clouds (CloudSEN12), and +1.5 mIoU over CloudSeg (official code) on its WHU-OPT-SAR protocol with one third of the parameters.
- Developed uncertainty-aware diffusion models for super-resolution by applying conformal prediction, wavelet transformation, and multi-modality alignment methods.
- Developed a unified multi-spectral super-resolution and wildfire segmentation pipeline for burnt-scar mapping on Sentinel-2 imagery.
- Preparing manuscripts on cloud-robust SAR–optical segmentation, uncertainty-aware super-resolution and wildfire segmentation for peer-reviewed conferences/journals.

Research Internship, Computer Vision and Mixed Reality Lab, RheinMain University of Applied Sciences, Germany June 2024 – Aug 2024

- Proposed an enhanced SWIN2SR transformer model with non-local sparse attention and high frequency preserving deep NN blocks for Single-Image Super-Resolution (SISR) of Sentinel-2 multispectral remote sensing images, achieving top LPIPS (0.3462) perceptual quality and 83% user preference while ranking second in PSNR (28.413 dB) on 4x SR tasks.
- Proposed a transformer-based Multi-Image Super-Resolution (MISR) by integrating non-local sparse attention into the encoding stage of transformer architecture, combined with wavelet-enhanced loss and random sequence shuffling, achieving state-of-the-art performance on the PROBA-V dataset (50.20 cPSNR, 0.9905 cSSIM).
- Two papers accepted at IEEE IGARSS (2025, 2026).

Research Assistant, TUKL NUST Research and Development Lab, NUST, Pakistan Sep 2023 – May 2024

- Developed Generative Adversarial Networks that generate forged offline signatures to test signature-verification systems against forgery attacks.

- Proposed a geometry-aware hybrid CNN-Transformer network integrating contourlet transform features, achieving state-of-the-art EERs of 6.84% (BHSig-H), 5.8% (BHSig-B), 0% (CEDAR), and 18.99% (UTSig) in writer-independent offline signature verification, with 3.6% FAR against GAN-generated fake signatures.
- Manuscript under review at IEEE Access.

Research Projects

Cloud-Robust SAR–Optical Semantic Segmentation from Clear-Sky Labels 2026 – Ongoing
(Supervisors: Prof. Dr. Ulrich Schwanecke, Prof. Dr. Faisal Shafait)

- Cloud-aware training of a two-stream SAR–optical network with a SAR-only head, tested on CloudSEN12 (real clouds), WHU-OPT-SAR, PIE-RGB-SAR and EarthMiss over 3 seeds; manuscript in preparation.

Multi-Spectral Super-Resolution and Wildfire Segmentation (Supervisor: Prof. Dr. Faisal Shafait) 2026 – Ongoing

- Unified multi-spectral super-resolution and burnt-scar segmentation pipeline on Sentinel-2 imagery of U.S. wildfires.

Industry Experience

Machine Learning Engineer, Dcube (US-based company), Islamabad, Pakistan Jan 2025 – Ongoing

- Developed and deployed an end-to-end system for real-time detection and recognition of truck license plates, driver faces, and phone usage, achieving 92.4% AP@50 on phone-usage detection and an average recognition accuracy of 85%.
- Led a team of 5 developers to distill and on-premise deploy a compact Italian sentence embedding model that retained 90–95% of teacher performance on MTEB retrieval, STS, clustering, and reranking benchmarks, while significantly reducing latency and inference cost for enterprise search and RAG workloads.

Start-up Incubation, National Incubation Center, University of Balochistan June 2021 – Dec 2022

- Developed an AI-based writing tool for novel writers [Code] and raised 600,000 PKR from the Pakistan Information Technology Board in exchange for 5% equity.

Technical Skills

Programming and DL: Python, C++, SQL, PyTorch, CUDA, Hugging Face, timm, MMSegmentation, OpenCV.

Remote Sensing: Sentinel-1/2, SAR, rasterio, GDAL, GeoPandas, QGIS, Sentinel Hub, Planetary Computer.

Engineering: Linux, Git, Docker, Conda, multi-GPU training, FastAPI, AWS (EC2, S3, Lambda), MongoDB, LaTeX.

Teaching and Leadership

Guest Lecturer (6 Cr. Hrs), CS-250: Data Structure and Algorithms, NUST SEECS Oct 2026

- Taught 6 credit hours of Prof. Dr. Faisal Shafait's course during his international travel: fundamentals of C++, trees and binary trees, and binary-tree traversal algorithms.

Computer Vision Lab Instructor, Advanced AI Bootcamp, NUST Oct 2023

- Designed and delivered lectures on advanced computer vision concepts, including object detection and image segmentation.
- Guided students in implementing deep learning algorithms for image classification and object detection.

Honours and Awards

- Selected (competitive) for the IEEE GRSS IADF School on Computer Vision for Earth Observation, Italy (Sep 2026)
- DAAD grant "Super Resolution of Satellite Image": research stay in Wiesbaden, Germany (summer 2024)
- Gold Medal for academic excellence, BS Software Engineering batch of 2017, BUITEMS (Jan 2022)
- Won the Pakistan Information Technology Board Start-up competition with funding of 600,000 PKR (June 2021)
- Awarded Balochistan Talent Award for innovative product idea (Dec 2020)
- Winner of the Tech Marathon coding competition organized by USAID (Sep 2019)

Professional References

Prof. Dr. Ulrich Schwanecke
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Prof. Dr. Faisal Shafait
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faisal.shafait@seecs.edu.pk